Slide Set #6

CLRM: Hypothesis Testing and Prediction

Pietro Coretto
pcoretto@unisa.it

Econometrics
Master in Economics and Finance (MEF)
Università degli Studi di Napoli "Federico II"

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Notes

Distributional assumption

Inference in finite sample is not distribution-free, we need to assume something which leads to well defined inference

Assumption (A5): Gaussian errors

$\varepsilon \mid \boldsymbol{X} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$

Remarks:

- (A5) is consistent with previous (A2), (A4)
- the distribution of $\varepsilon \mid \boldsymbol{X}$ does not depend on $\boldsymbol{X}$

Notes

Proposition: distribution of $\boldsymbol{b}$

Under (A1)-(A5)

$$(\boldsymbol{b}-\boldsymbol{\beta})\,|\,\boldsymbol{X} \sim \mathcal{N}\left(\boldsymbol{0}, \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\right)$$

Proof. From the estimation error decomposition

$$(\boldsymbol{b}-\boldsymbol{\beta}) = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon} = \boldsymbol{A}\boldsymbol{\varepsilon}, \qquad \text{where} \quad \boldsymbol{A} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'.$$

By (A5) plus the linearity property of the multivariate normal, $(\boldsymbol{b}-\boldsymbol{\beta})\,|\,\boldsymbol{X}$ is normally distributed with parameters

$$\mathrm{E}[(\boldsymbol{b}-\boldsymbol{\beta})\,|\,\boldsymbol{X}] = \mathrm{E}[\boldsymbol{A}\boldsymbol{\varepsilon}\,|\,\boldsymbol{X}] = \boldsymbol{A}\,\mathrm{E}[\boldsymbol{\varepsilon}\,|\,\boldsymbol{X}] = \boldsymbol{0}$$

$$\mathrm{Var}[(\boldsymbol{b}-\boldsymbol{\beta})\,|\,\boldsymbol{X}] \;=\; \mathrm{Var}[\boldsymbol{A}\boldsymbol{\varepsilon}\,|\,\boldsymbol{X}] \;=\; \boldsymbol{A}\,\mathrm{Var}[\boldsymbol{\varepsilon}\,|\,\boldsymbol{X}]\boldsymbol{A}'$$

$$\begin{aligned}\boldsymbol{A}\,\mathrm{Var}[\boldsymbol{\varepsilon}\,|\,\boldsymbol{X}]\boldsymbol{A}' =& (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'(\sigma^2\boldsymbol{I})((\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}')' \\ =& \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} \\ =& \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\end{aligned}$$

which completes the proof ∎

Notes

Important property of the multivariate normal distribution: all its marginals are also normally distributed.

Therefore, all the coordinates of the vector $(b_k - \beta_k)$ have a univariate normal distribution

$$(b_k-\beta_k)\,|\,\boldsymbol{X} \sim \mathcal{N}\left(0, \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}_{kk}\right)$$

These distributional results can be used to make inference on $\boldsymbol{\beta}$

Notes

General remark on testing

Classical hypothesis testing (aka Neyman–Pearson optimality):

- Fix $\alpha = \Pr\{\text{false rejection}\} = \Pr\{\text{Type-1 error}\}$
- Max Power $=$ Max $\Pr\{\text{correct rejection}\} =$ Max $\Pr\{\overline{\text{Type-2 error}}\}$

Calculations of $\Pr\{\cdot\}$ requires knowledge of probability laws for underlying data generating process.

We always test more than just H_0-vs-H_1, because in reality we calculate

- $\alpha = \Pr\{\text{reject } \mathsf{H}_0 \,|\, (\mathsf{H}_0 \text{ is true}) \;\&\; (\text{fullfiled model assumptions})\}$
- Power$=\Pr\{\text{reject } \mathsf{H}_0 \,|\, (\mathsf{H}_1 \text{ is true}) \;\&\; (\text{fulfilled model assumptions})\}$

The following testing decision rules are optimal in the sense of Neyman–Pearson only if the underlying data generating process fulfills all/some of A1–A5.

Notes

Testing individual regression coefficients

Tests for individual parameters

$$\mathsf{H}_0 : \beta_k = \beta_k^*$$

t-ratio or *t-test* statistic

$$t_k = \frac{b_k - \beta_k^*}{\sqrt{s^2(\boldsymbol{X}'\boldsymbol{X})^{-1}_{kk}}} = \frac{b_k - \beta_k^*}{\mathrm{Se}(b_k)}$$

It can be shown that

$$t_k \,|\, \boldsymbol{X} \overset{\mathsf{H}_0}{\sim} \mathsf{T}(n-K)$$

Notes

Default test

$$\begin{cases} \mathsf{H}_0 : \beta_k = 0 \\ \mathsf{H}_1 : \beta_k \neq 0 \end{cases}$$

Decision rule at significance level α:

$$\text{reject } \mathsf{H}_0 \text{ if } |t_k| > q_{1-\frac{\alpha}{2}}$$

where $q_{1-\frac{\alpha}{2}}$ is the $(1-\frac{\alpha}{2})$-quantile of the $\mathsf{T}(n-K)$

Confidence interval

Let $(1-\alpha)$ be the confidence level. The confidence interval for β_k is given by

$$b_k \ \pm \ q_{1-\frac{\alpha}{2}} \, \mathrm{Se}(b_k)$$

Notes

Interlude: nested and non-nested models

One common approach to testing a hypothesis is to formulate a model that contains the hypothesis as a restriction on its parameters.

Investment model M1: *investors cares about I_t=investment, i_t = the nominal interest rate, Δp_t = inflation, Y_t = real output*

$$\ln I_t = \beta_1 + \beta_2 i_t + \beta_3 \Delta p_t + \beta_4 \ln Y_t + \varepsilon_t$$

Where the four parameters are *unrestricted*: $\beta_1, \beta_2, \beta_3, \beta_4$ can do whatever in this model formulation

Notes

Investment model M2: *investors only care about real interest rates and real output*

$$\ln I_t = \gamma_1 + \gamma_2(i_t - \Delta p_t) + \gamma_4 \ln Y_t + \varepsilon_t$$

M2 = (M1 + restriction): in fact taking M1 and assuming that $\beta_2 = -\beta_3$ we get M2. We say that *M2 is nested in M1*.

Model B is nested in model A:

- both A and B have regression functions defined on the same space (same regressors),
- parameter space of B can be written as a restricted version of the parameter space of A

Notes

Investment model M3: *investors care only about the nominal interest rate and real output*

$$\ln I_t = \theta_1 + \theta_2 i_t + \theta_3 \ln Y_t + \varepsilon_t$$

Investment model M4: *investors care only about inflation and real output*

$$\ln I_t = \eta_1 + \eta_3 \Delta p_t + \eta_3 \ln Y_t + \varepsilon_t$$

Note that also M3 and M4 are special cases of M1, however there is no nesting here. In fact, the regression function of M1, M3, and M4 map different spaces.

⚠ It's true that you can obtain M3 by setting $\beta_3 = 0$ in M1, but this happens for just one specific value of $\beta_3 = 0$. Setting $\beta_3 = 0$ in M1 changes the effective dimension of the regression space from $K = 4$ to $K = 3$.

Notes

Testing linear hypothesis

Many tests of interest can be written in terms of a linear system. Suppose we have the linear model

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \text{error}$$

We want to test the joint hypothesis that $\beta_1 = \beta_2 = \beta_3 = 0$. This can be formulated as a system of 3 equations: $\{\beta_1 = 0, \beta_2 = 0, \beta_3 = 0\}$ For an appropriate choice of a (3×5)-matrix $\boldsymbol{R}$, and a (3×1)-column vector $\boldsymbol{r}$, we write the null hypothesis in matrix form as

$$\boldsymbol{R\beta} = \boldsymbol{r} \quad \text{that is} \quad \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{that is} \begin{cases} \beta_1 = 0 \\ \beta_2 = 0 \\ \beta_3 = 0 \end{cases}$$

Notes

We consider a set of J linear restrictions of the form

$$\mathsf{H}_0 : \boldsymbol{R\beta} = \boldsymbol{r}$$

where $\boldsymbol{R} = (R_{ij})$ is a $(J \times K)$ matrix, and $\boldsymbol{r}$ is a $(J \times 1)$ column vector

That is, under the null hypotesis is true that

$$\begin{cases} R_{11}\beta_1 + R_{12}\beta_2 + \cdots + R_{1K}\beta_K = & r_1 \\ R_{21}\beta_1 + R_{22}\beta_2 + \cdots + R_{2K}\beta_K = & r_2 \\ \qquad\qquad \vdots & \\ R_{J1}\beta_1 + R_{J2}\beta_2 + \cdots + R_{JK}\beta_K = & r_J \end{cases}$$

Assume $J < K$, and that $\boldsymbol{R}$ is full row-rank, this means that *rows of* $\boldsymbol{R}$ *are linearly independent*.

Notes

Strategies:

Wald-type testing approach

- estimate the unrestricted model
- based on unrestricted estimates construct a test-statistic that measures how close to the restricted case we are

LRT-type testing approach

- estimate the unrestricted model
- estimate the restricted model
- construct a test-statistic that measures the loss of fit

Notes

Wald's type testing

H_0 is likely to hold when $(\boldsymbol{Rb} - \boldsymbol{r})$ is small enough, or close to $\mathbf{0}$. Hence, under H_0 we expect the norm $\|\boldsymbol{Rb} - \boldsymbol{r}\|$ to be small

The components of $\boldsymbol{b}$ are usually highly correlated, so it does $\boldsymbol{Rb}$. Therefore, the Euclidean norm is not appropriate.

We need an *orthogonalization* device, namely

$$(\boldsymbol{Rb} - \boldsymbol{r})'\boldsymbol{Q}^{-1}(\boldsymbol{Rb} - \boldsymbol{r})$$

that is the Mahalanobis distance of $\boldsymbol{Rb}$ from $\boldsymbol{r}$ for an appropriate scaling matrix $\boldsymbol{Q} = \mathrm{Var}[\boldsymbol{Rb}]$

This kind of testing strategy is called *Wald-test*. It is much more general than this particular case

Notes

The popular F-statistic, in the context of multiple regression, is nothing more than a Wald-statistics.

Consider the *F-ratio*

$$F = \frac{(\boldsymbol{Rb} - \boldsymbol{r})' \left[\boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}' \right]^{-1} (\boldsymbol{Rb} - \boldsymbol{r}) \ / \ J}{s^2}$$

Let $\boldsymbol{Q}^{-1} = \left[s^2 \boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}' \right]^{-1}$, then

$$F = \frac{1}{J}(\boldsymbol{Rb} - \boldsymbol{r})' \boldsymbol{Q}^{-1} (\boldsymbol{Rb} - \boldsymbol{r})$$

We will show that under H_0, $\boldsymbol{Q}$ is in fact an estimate of $\mathrm{Var}[\boldsymbol{Rb} - \boldsymbol{r}]$

Notes

Proposition: distribution of the Wald's statistics

Assume A1–A5 and H_0, then $F \mid \boldsymbol{X} \overset{\mathsf{H}_0}{\sim} \mathsf{F}(J, n-K)$

Proof. Under H_0 $\boldsymbol{r} = \boldsymbol{R}\boldsymbol{\beta}$, therefore

$$\boldsymbol{Rb} - \boldsymbol{r} = \boldsymbol{Rb} - \boldsymbol{R}\boldsymbol{\beta} = \boldsymbol{R}(\boldsymbol{b} - \boldsymbol{\beta})$$

Moreover since $(\boldsymbol{b} - \boldsymbol{\beta})$ is normally distributed

$$(\boldsymbol{Rb} - \boldsymbol{r}) \mid \boldsymbol{X} \overset{\mathsf{H}_0}{\sim} \boldsymbol{R}\, \mathcal{N}\left(\boldsymbol{0}, \sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}\right)$$

Which becomes

$$(\boldsymbol{Rb} - \boldsymbol{r}) \mid \boldsymbol{X} \sim \mathcal{N}\left(\boldsymbol{0}, \sigma^2 \boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}'\right)$$

Then

$$(\boldsymbol{Rb} - \boldsymbol{r})' \left[\sigma^2 \boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}' \right]^{-1} (\boldsymbol{Rb} - \boldsymbol{r}) \sim \chi^2(J)$$

Notes

By A5

$$\frac{\boldsymbol{\varepsilon}}{\sigma} \,\Big|\, \boldsymbol{X} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_n)$$

Consider the residual maker matrix $\boldsymbol{M}$, since $\operatorname{rank}(\boldsymbol{M}) = n - K$

$$\left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right)' \boldsymbol{M} \left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right) \,\Big|\, \boldsymbol{X} = \mathcal{N}(\mathbf{0}, \boldsymbol{I}_n)' \boldsymbol{M}\, \mathcal{N}(\mathbf{0}, \boldsymbol{I}_n) \sim \chi^2(n-K)$$

Now recall that we can write $\boldsymbol{e}'\boldsymbol{e} = \boldsymbol{\varepsilon}'\boldsymbol{M}\boldsymbol{\varepsilon}$, therefore

$$\frac{\boldsymbol{e}'\boldsymbol{e}}{\sigma^2} \sim \chi^2(n-K)$$

And

$$\frac{s^2}{\sigma^2} = \frac{1}{\sigma^2}\frac{\boldsymbol{e}'\boldsymbol{e}}{n-k} \sim \frac{\chi^2(n-K)}{n-K}$$

Rewrite F as

$$F = \frac{(\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r})' \left[\sigma^2 \boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}'\right]^{-1} (\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r}) \;/\; J}{\frac{s^2}{\sigma^2}}$$

Notes

Multiply and divide F by $\frac{1}{\sigma^2}$

$$F = \frac{(\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r})' \left[\sigma^2 \boldsymbol{R}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}'\right]^{-1} (\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r}) \;/\; J}{\frac{s^2}{\sigma^2}}$$

Then

$$F \,|\, \boldsymbol{X} = \frac{\frac{\chi^2(J)}{J}}{\frac{\chi^2(n-K)}{n-K}} \overset{\mathsf{H}_0}{\sim} \mathsf{F}(J, n-K)$$

because numerator and denominator are independent given $\boldsymbol{X}$, in fact their conditional distributions do not depend on $\boldsymbol{X}$ ■

Notes

Test

$$\begin{cases} \mathsf{H}_0 : \boldsymbol{R}\beta = \boldsymbol{r} \\ \mathsf{H}_1 : \boldsymbol{R}\beta \neq \boldsymbol{r} \end{cases}$$

Decision rule at significance level α:

$$\text{reject } \mathsf{H}_0 \text{ if } f > q_{1-\alpha}$$

where

- f is the observed value of the F-statitic
- $q_{1-\alpha}$ is the $(1-\alpha)$-quantile of the $\mathsf{F}(J, n-K)$ distribution

Notes

Prediction

Suppose we want to predict Y knowing that the regressor vector X is equal to some value $\boldsymbol{x}^0$.

The optimal predictor theorem (OPT) tells us that the best prediction (in the MSE sense) is $\mathrm{E}[Y \mid X = \boldsymbol{x}^0]$, an estimate for this predictor is given by the estimated regression function at $\boldsymbol{x}^0$

$$\widehat{\mathrm{E}[Y \mid X = \boldsymbol{x}^0]} = \hat{y}^0 = \boldsymbol{b}'\boldsymbol{x}^0$$

The OPT says that this is the minimum MSE predictor. Since it is unbiased (why?) $\implies$ it's the minimum variance predictor!

The prediction error is given by $\hat{y}^0 - y^0$ where

$$y^0 = \boldsymbol{\beta}'\boldsymbol{x}^0 + \varepsilon^0 \qquad \text{and} \qquad \hat{y}^0 = \boldsymbol{b}'\boldsymbol{x}^0$$

Therefore

$$e^0 = \hat{y}^0 - y^0 = (\boldsymbol{b} - \boldsymbol{\beta})'\boldsymbol{x}^0 - \varepsilon^0$$

Notes

Proposition: bias/variance of the prediction error

Assume A1–A4. Let e^0 be the prediction error of the the optimal linear predictor at a point $\boldsymbol{x}^0$, then

1. $\mathrm{E}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] = 0$
2. $\mathrm{Var}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] = \boldsymbol{x}^{0\prime} \left(\sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}\right) \boldsymbol{x}^0 + \sigma^2$

Proof.

$$\begin{aligned}
\mathrm{E}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] &= \mathrm{E}[(\boldsymbol{b} - \boldsymbol{\beta})'\boldsymbol{x}^0 - \varepsilon^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] \\
&= \mathrm{E}[(\boldsymbol{b} - \boldsymbol{\beta})' \mid \boldsymbol{X}, \boldsymbol{x}^0]\boldsymbol{x}^0 - \mathrm{E}[\varepsilon^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] \\
&= 0
\end{aligned}$$

$$\begin{aligned}
\mathrm{Var}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] &= \mathrm{Var}[(\boldsymbol{b} - \boldsymbol{\beta})'\boldsymbol{x}^0 - \varepsilon^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] \\
&= \mathrm{Var}[(\boldsymbol{b} - \boldsymbol{\beta})'\boldsymbol{x}^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] + \mathrm{Var}[\varepsilon^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] \\
&= \mathrm{Var}[(\boldsymbol{b} - \boldsymbol{\beta})'\boldsymbol{x}^0 \mid \boldsymbol{X}, \boldsymbol{x}^0] + \sigma^2 \\
&= \boldsymbol{x}^{0\prime} \left(\sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}\right) \boldsymbol{x}^0 + \sigma^2
\end{aligned}$$

Notes

$\mathrm{Var}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0]$ can be estimated with s^2 replacing σ^2.

$\mathrm{Se}(e^0) = \sqrt{\mathrm{Var}[e^0 \mid \boldsymbol{X}, \boldsymbol{x}^0]}$ is the standard error of the prediction error

Prediction confidence interval

Consider the prediction at $\boldsymbol{x}^0$. The $(1-\alpha)$ confidence interval for the predicted value is

$$\hat{y}^0 \pm q_{(1-\frac{\alpha}{2})} \mathrm{Se}(e^0)$$

where $q_{1-\frac{\alpha}{2}}$ is the $(1-\frac{\alpha}{2})$-quantile of the $\mathsf{T}(n-K)$ distribution.

Notes